

Regional HPC - Laguna

Who We Are

SoCal RC&D Alliance is a consortium of 12 academic institutions in Southern California dedicated to fostering multidisciplinary, multi-institutional research collaborations and educational opportunities. The Alliance provides a high-speed network connecting participating universities and access to advanced research cyberinfrastructure.

At the core of this effort is Laguna, a state-of-the-art HPC system hosted at the University of Southern California (USC) and managed by the Center for Advanced Research Computing (CARC). Laguna features 16 compute nodes (4096 cores), and 8 dual GPU nodes (16 L40s), offering powerful computational resources to advance research across various disciplines.

Our Partners



Contact Us

laguna-support@usc.edu



uschpc.github.io/regional-computing-website/



Computing



Data



Education



Research and Projects

Get Started

Request Access

Contact us at the Laguna support email (see below) or reach out to the research support staff at your institution's local IT department. They will provide you with a form to fill out requesting a project on Laguna.

Project Setup

Staff at CARC receive the request form and provision your account in the Laguna user portal. Log in to the portal and update the required fields with your project information.

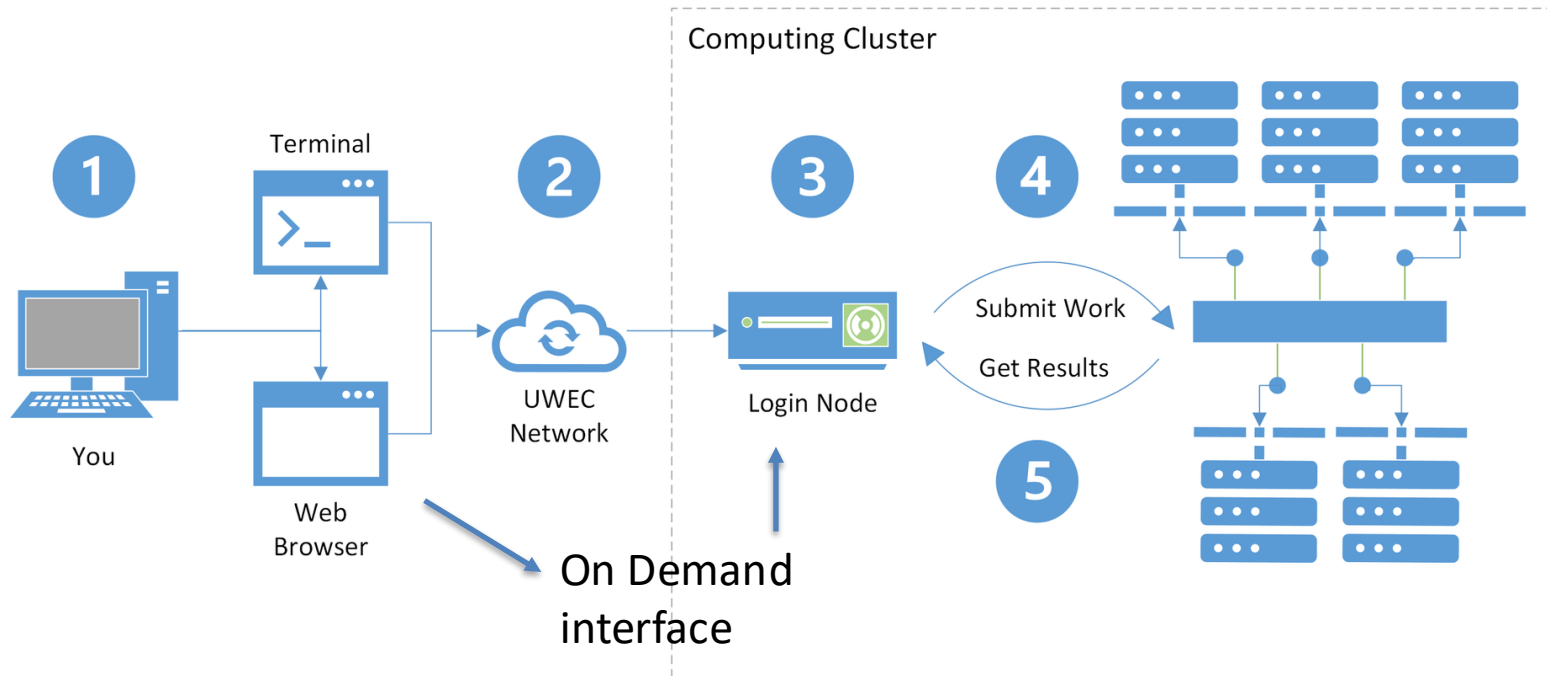
System Access

SSH into the login node or log in using Laguna OnDemand to get started running jobs.

Funded By



HPC Structure



CARC's OnDemand service provides users

- graphical, browser-based access to the Discovery and Endeavour HPC clusters and their /home1, /project2, /cryoem2, and /scratch1 directories. OnDemand offers:
 - Easy file management
 - Command line shell access
 - Slurm job submission and management
 - Interactive applications, including Jupyter notebooks and RStudio Server

Use cases

- OnDemand has a number of use cases that are not available with the traditional command-line interface to HPC resources:
 - File management with a graphical interface
 - Text editing with a graphical interface
 - Shell access with a web browser
 - Slurm job submission with a graphical interface
 - Interactive use of graphical apps
 - Development of code and notebooks with JupyterLab
 - Development of R code and notebooks with RStudio Server
 - Development of code with VS Code
 - High-performance and interactive data visualization

Sensitive data

- Currently, CARC systems **do not** support the use or storage of sensitive data. If your research work includes sensitive data, including but not limited to HIPAA-, FERPA-, or CUI-regulated data, see our [Secure Computing Compliance Overview](#) or contact us at carc-support@usc.edu before using our systems.

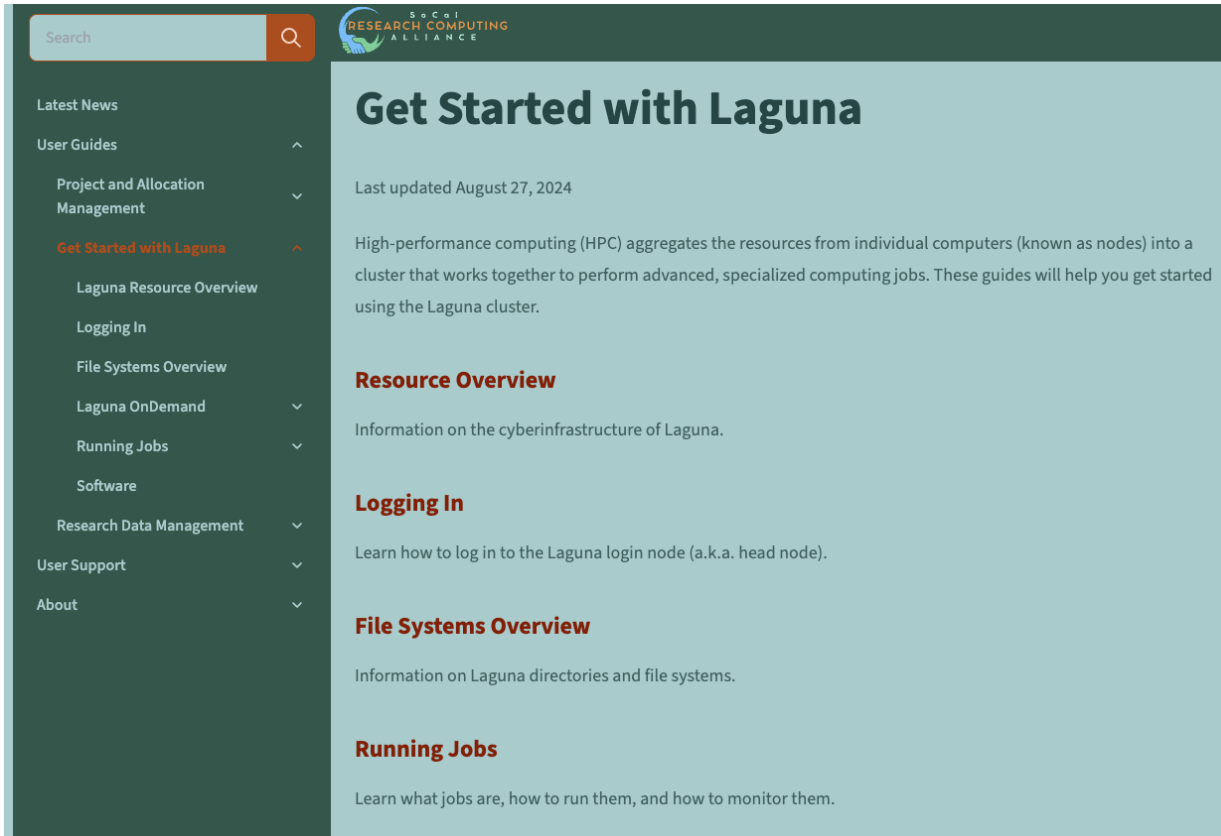
PRE-REQUISITES

- Each PI should contact their local administrator to request for Laguna access. Administrators will provide PIs with a Google form to fill out.
- We need your eduPersonPrincipalName (EPPN) and email address for provisioning. Your EPPN will be your Laguna username.
 - Go to our test page to find the values (may not be what you expect)
 - <https://hpcaccount.usc.edu/static/secure/incommon.php>
- CARC support will provision the project and PI account.
- Once the project is created, PIs can access it in the user portal.

Workflow

- The workflow for using Discovery typically consists of the following steps:
 1. Connecting to the USC network or VPN
 2. Logging in to the login node
 3. Organizing files
 4. Transferring files
 5. Installing and running software
 6. Testing your job interactively on a compute node
 7. Submitting your job to the job scheduler to run it remotely on a compute node
 8. Monitoring your job and checking the results when it has completed

Resources



The screenshot shows the website of the Regional Computing Alliance. The header includes a search bar and the logo. The left sidebar contains a navigation menu with categories like 'Latest News', 'User Guides', 'Project and Allocation Management', 'Get Started with Laguna' (highlighted), 'Laguna Resource Overview', 'Logging In', 'File Systems Overview', 'Laguna OnDemand', 'Running Jobs', 'Software', 'Research Data Management', 'User Support', and 'About'. The main content area is titled 'Get Started with Laguna' and includes a date 'Last updated August 27, 2024'. It provides an overview of High-performance computing (HPC) and lists several guides: 'Resource Overview', 'Logging In', 'File Systems Overview', and 'Running Jobs'.

Get Started with Laguna

Last updated August 27, 2024

High-performance computing (HPC) aggregates the resources from individual computers (known as nodes) into a cluster that works together to perform advanced, specialized computing jobs. These guides will help you get started using the Laguna cluster.

Resource Overview

Information on the cyberinfrastructure of Laguna.

Logging In

Learn how to log in to the Laguna login node (a.k.a. head node).

File Systems Overview

Information on Laguna directories and file systems.

Running Jobs

Learn what jobs are, how to run them, and how to monitor them.

<https://uschpc.github.io/regional-computing-website/user-guides/get-started-laguna.html>

Resources

Search

USC | Center for Advanced Research Computing
Enabling scientific breakthroughs at scale

CARC OnDemand Overview

Last updated July 12, 2023

CARC's OnDemand service provides users graphical, browser-based access to the Discovery and Endeavour HPC clusters and their /home1, /project2, /cryoem2, and /scratch1 directories. OnDemand offers:

- Easy file management
- Command line shell access
- Slurm job submission and management
- Interactive applications, including Jupyter notebooks and RStudio Server

OnDemand is available to all CARC users. To access OnDemand, you must belong to an active project in the [CARC user portal](#).

CARC OnDemand will only be accessible via a connection to either USC's Secure network or a USC VPN. Instructions for setting up a VPN connection can be found at the following links:

- [Windows](#)
- [MacOS](#)
- [Linux](#)

[Log in to CARC OnDemand](#)

We recommend using OnDemand in a private browser to avoid potential permissions issues related to your browser's

Table of Contents

- [Use cases](#)
- [Additional resources](#)

<https://www.carc.usc.edu/user-guides/carc-ondemand/ondemand-overview>

Hands on:

Access your HPC account and find your project and allocation

 **USC** | Advanced Research Computing
Enabling scientific breakthroughs at scale

Home Center Summary Project ▾ Help

danperez@cmc.edu ▾

Your Federated eppn (eduPersonPrincipalName) is: **danperez@cmc.edu**. Pass this info to your PI to add you to their project for access to USC CARC Regional cluster.

Projects »

Project Title	PI
 Please update	danperez@cmc.edu
 CMC_Onboarding	JehoPark@cmc.edu

Allocations »

Project	Information	Quota	Status
Please update	slurm_account: danperez_1422 Open OnDemand - Laguna		 Active
CMC_Onboarding	slurm_account: jehopark_1421 Open OnDemand - Laguna		 Active

CARC Cluster Status

[More status info](#)
[Detailed Job info](#)

<https://hpcaccount.usc.edu/>



Public and Private keys

Generating a key pair provides you with two long string of characters: a public and a private key. You can place the public key on any server and then unlock it by connecting to it with a client that already has the private key. When the two match up, the system unlocks without the need for a password. You can increase security even more by protecting the private key with a passphrase.

Create an RSA key pair

- To create the key pair on your personal computer, open a shell session and follow the instructions:

```
$ ssh-keygen -t rsa -b 4096
```

- This command prompts you to designate a location for the keys to be saved:

```
Enter file in which to save the key (/home/user_id/.ssh/id_rsa):
```

You can press enter here, saving the file to the default directory (e.g., /home/user_id/.ssh/id_rsa on macOS or Linux or C:\Users\user_id\.ssh\id_rsa on Windows).

- Next is the passphrase prompt, provide a password for enhanced security:

```
Enter passphrase (empty for no passphrase):
```

The entire key generation process looks something like this:

```
$ ssh-keygen -t rsa -b 4096
Generating public/private rsa key pair.
Enter file in which to save the key (/home/user_id/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/user_id/.ssh/id_rsa.
Your public key has been saved in /home/user_id/.ssh/id_rsa.pub.
The key fingerprint is:
4a:dd:0a:c6:35:4e:3f:ed:27:38:8c:74:44:4d:93:67 user_id@local
The key's randomart image is:
+--[ RSA 4096 ]-----+
|           .oo.      |
|           . o.E     |
|          + . o      |
|         . = = .     |
|          = S = .    |
|        o + = +      |
|         . o + o .   |
|           . o       |
|                     |
+-----+

```

Hands On: Add public key to user portal

Once the keys are generated, copy the public key.

- Mac OS:

```
cat ~/.ssh/id_rsa.pub | pbcopy
```

- Windows:

```
cat ~/.ssh/id_rsa.pub
```

- Navigate to the user portal and select your username in the top right corner to display the drop down menu. Select User Profile and paste your SSH key into the public SSH key field:

The screenshot shows the USC Advanced Research Computing user portal. The top navigation bar includes the USC logo, the text "Advanced Research Computing" and "Enabling scientific breakthroughs at scale", and links for Home, Center Summary, Project, Admin, Invoice, and Help. The user's name "dcannell" is displayed in the top right corner with a dropdown menu containing "User Profile" and "Log Out". Below the navigation bar, there are links for "Projects »" and "Allocations »". The main content area is titled "User Profile" and shows the user's name "CARC". The "University Role(s):" field is empty. The "Email:" field contains "Daniel.Perez@ClaremontMcK". The "PI Status:" field has a green checkmark. The "Public SSH Key (Regional cluster users only):" field is highlighted with a red box and contains the following text: "ssh-rsa AAAAB3NzaC1yc2EAAAADA...". A red box also highlights the "Update SSH Key" button at the bottom of the field.

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Home Center Summary Project Admin Invoice Help

dcannell

User Profile

Log Out

Projects » Allocations »

User Profile

CARC

University Role(s):

Email: Daniel.Perez@ClaremontMcK

PI Status: ☒

Public SSH Key (Regional cluster users only):

1 ssh-rsa
AAAAB3NzaC1yc2EAAAADA
v4XiZYR6oNz/pzShXytNUhY
K9IxsLJFZ8ezDukhINCLwh
yL3jotWLRK5pDdyL04q+Wa
6L9vaU3tnDAlyxpQPhGF0o
Yd8YVFtaelOQO4B1VqhAxG
Q1/2JdgdIMa3C4aQKct2/3z

2 Update SSH Key

Laguna OnDemand

<https://laguna-ood.carc.usc.edu/pun/sys/dashboard/>

USC CARC Laguna Apps Files Jobs Clusters Interactive Apps My Interactive Sessions



Advanced Research Computing
Enabling scientific breakthroughs at scale

OnDemand provides an integrated, single access point for all of your HPC resources.

New users, see our [Getting Started with CARC OnDemand Guide](#).

Pinned Apps A featured subset of all available apps



JupyterLab
System Installed App



RStudio Server
System Installed App



Active Jobs
System Installed App



Home Directory
System Installed App

Recently Used Apps



JupyterLab
System Installed App



RStudio Server
System Installed App

Interactive Apps

- We currently offer [JupyterLab](#), [RStudio Server](#) for Laguna
- Other applications, such as [Stata](#), [Mathematica](#), and [MATLAB](#), are available on the Traveler Desktop under the Interactive Apps dropdown (Not Laguna).

Running Jobs with CARC OnDemand

- OnDemand provides two related job management tools. The Active Jobs tool allows you to monitor queued and running jobs and the Job Composer tool allows you to create and submit jobs via your web browser:
- The Active Jobs tool will show you all jobs currently in the queue (running or queued), regardless of how the jobs were submitted.
- Selecting Job Composer in the Jobs menu will open the Jobs page, where you can create new jobs and submit them to the cluster, and inspect the results of jobs submitted via this tool.

Interactive Apps

Running sessions

Interactive Apps

Desktops

Laguna Desktop

Servers

Code Server

JupyterLab

RStudio Server

JupyterLab (1872) 1 node | 1 core | Running

Host: `>_ c22-01.carc.usc.edu` Cancel

Created at: 2025-02-05 22:32:58 PST

Time Remaining: 57 minutes

Session ID: `821272b1-14e8-474c-9f11-c43387227ea5`

Problems with this session? [Submit support ticket](#)

Connect to JupyterLab

Laguna Desktop (1871) 1 node | 1 core | Running

Host: `>_ c22-01.carc.usc.edu` Cancel

Created at: 2025-02-05 22:32:38 PST

Time Remaining: 56 minutes

Session ID: `cdc30d2b-4f97-4136-9bd9-04847364abd2`

Problems with this session? [Submit support ticket](#)

Compression

0 (low) to 9 (high)

Image Quality

0 (low) to 9 (high)

Launch Laguna Desktop View Only (Share-able Link)



File System



Home



Sign in to RStudio

Username:

Password:

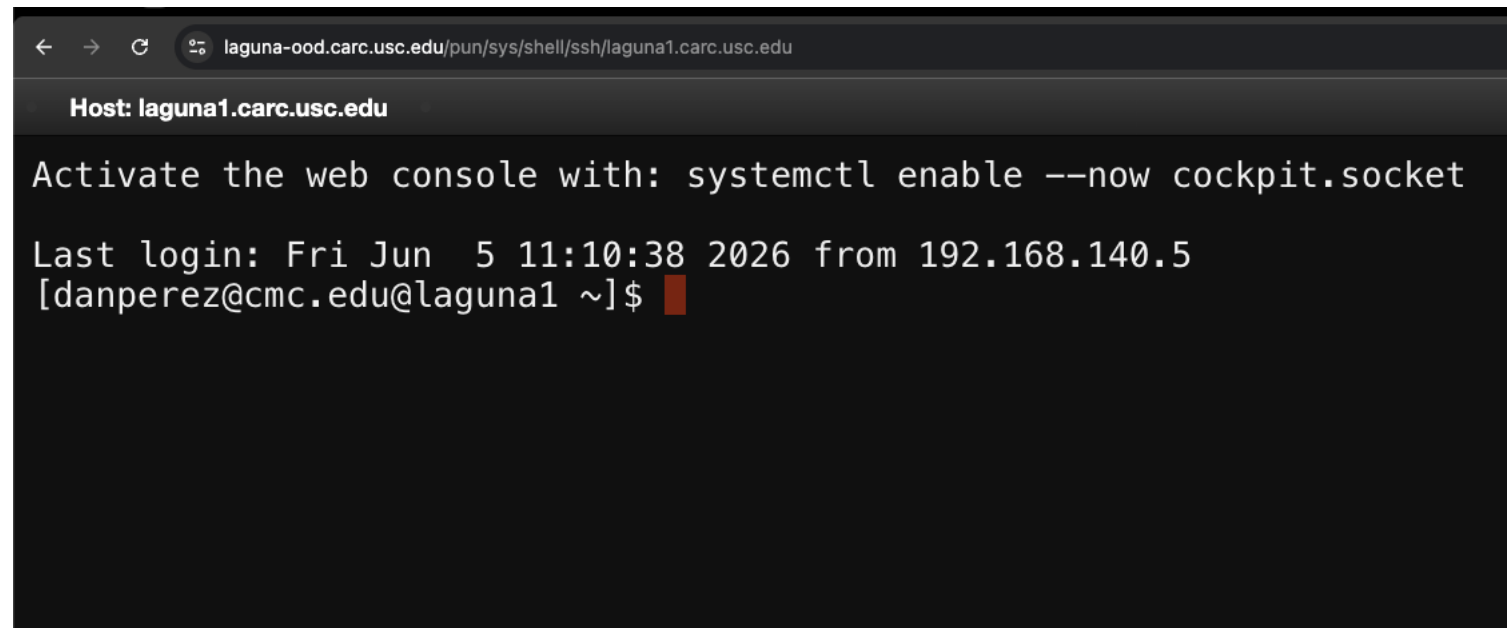
☐ Stay signed in when browser closes

You will automatically be signed out after 60 minutes of inactivity.

Sign in

Shell Access

OnDemand provides the ability to SSH into the Laguna Cluster



The screenshot shows a web browser window with the address bar displaying `laguna-ood.carc.usc.edu/pun/sys/shell/ssh/laguna1.carc.usc.edu`. Below the address bar, the host is identified as `Host: laguna1.carc.usc.edu`. The terminal content includes the instruction `Activate the web console with: systemctl enable --now cockpit.socket`, the login history `Last login: Fri Jun 5 11:10:38 2026 from 192.168.140.5`, and the current shell prompt `[danperez@cmc.edu@laguna1 ~]$` followed by a red cursor.

```
← → ↻ laguna-ood.carc.usc.edu/pun/sys/shell/ssh/laguna1.carc.usc.edu
Host: laguna1.carc.usc.edu
Activate the web console with: systemctl enable --now cockpit.socket
Last login: Fri Jun 5 11:10:38 2026 from 192.168.140.5
[danperez@cmc.edu@laguna1 ~]$
```

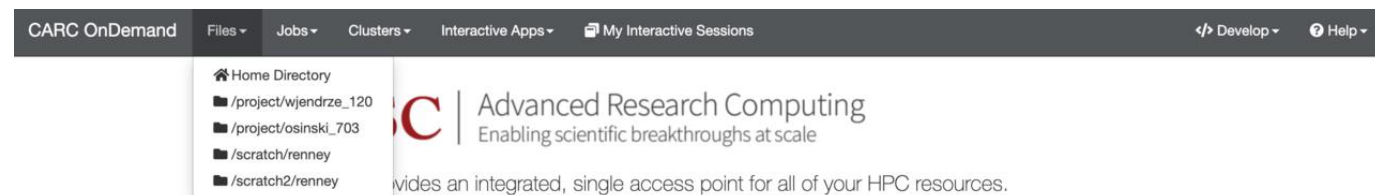
Hands on:

<https://laguna-ood.carc.usc.edu/pun/sys/dashboard/>

- Access to On Demand and create three sessions:
 - Linux visual interface
 - Jupyter Notebook / R session
 - Shell access

Data Management

- OnDemand provides a web-based file explorer that can be used to upload, download, copy, delete, rename, and edit data. OnDemand has quick links to your:
 - CARC home directory
 - CARC project directory(ies)
 - CARC cryo-EM storage directory
 - CARC scratch directories



Data Management (Cont.)

- Every user will have access to their /home1 and /scratch1 directories.
- If you have access to a project space in a project directory, you will also have access to this directory.
- The project directory's name will be of the form /project2/<PI_username>_<Project_ID>
- If you have access to cryo-EM storage, the directory's name will be of the form /cryoem2/<PI_username>_<Project_ID>

Uploading and downloading files

- OnDemand provides a basic file upload/download feature. However, this should be reserved for relatively small files (e.g., scripts and log files). For larger files, use one of the applications described in our transfer guides.
- There are several ways to upload files. The first is to simply navigate to the desired destination directory in OnDemand, and then drag the file from your local computer to the OnDemand browser window. You can also click the “Upload” button to open a dialog that will allow you to navigate your local computer and select which files you want to upload:
- To download files, select the file(s) in the OnDemand directory that you wish to download, and then use the blue Download button. You can select multiple files by holding down the Shift or Ctrl (or Command on a Mac) key as you select the files. If you select multiple files or a folder, OnDemand will zip the files and download a single .zip archive to your computer.

Transferring files

Different ways of transferring a file:

- CARC GUI
- Globus
- Terminal (SSH connection)

Hands on:

- Copy some files to your project directory using the graphical interface

USC CARC Laguna

Apps ▾ Files ▾ Jobs ▾ Clusters ▾ Interactive Apps ▾ My Interactive Sessions

Help ▾

Logged in as danperez@cmc.edu

Log Out

Open in Terminal ▾ Refresh + New File New Directory Upload Download Copy/Move Delete

Home Directory

/project/JehoPark_1421

/project/danperez_1422

/project/JehoPark_1642

↑ / home1 / danperez@cmc.edu / Change directory Copy path

☐ Show Owner/Mode ☐ Show Dotfiles Filter:

Showing 6 of 27 rows - 0 rows selected

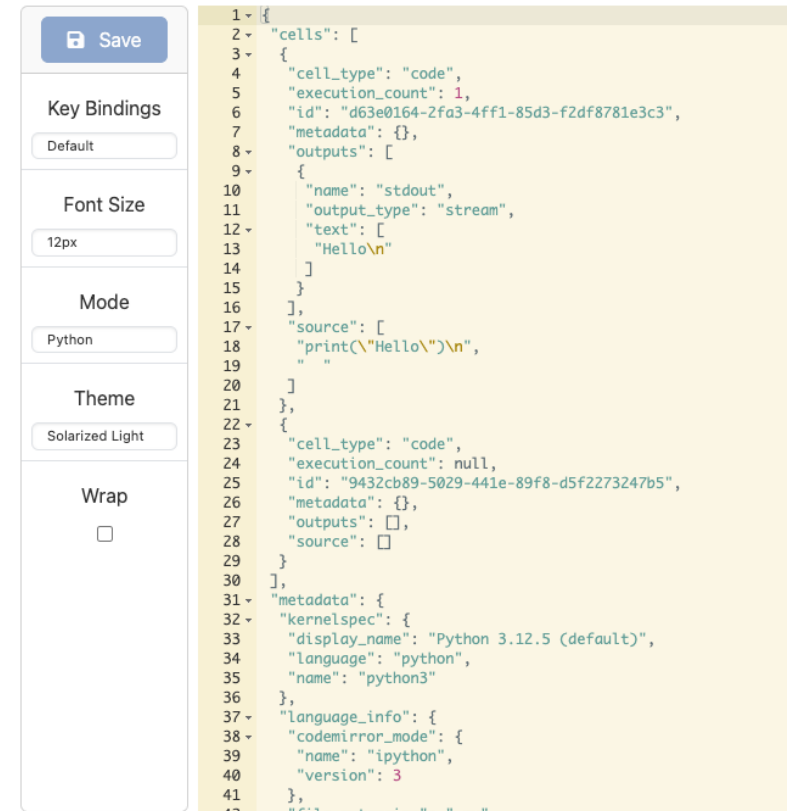
☐	Type ▲	Name		Size	Modified at
☐	Folder	Desktop	⋮ ▾	-	2/5/2025 10:32:43 PM
☐	Folder	ondemand	⋮ ▾	-	9/30/2024 11:14:15 AM
☐	Folder	slurm-tools	⋮ ▾	-	4/15/2025 10:38:56 AM
☐	File	Daniel_test.ipynb	⋮ ▾	6.41 kB	9/30/2024 11:59:18 AM
☐	File	test	⋮ ▾	5.00 B	3/6/2025 11:19:30 AM
☐	File	Untitled.ipynb	⋮ ▾	926.00 B	2/5/2025 10:37:09 PM

File operations

- Most file operations can be accomplished by selecting a file in the File Explorer and choosing the desired operation from the main menu:
- You have the ability to perform the following file operations in OnDemand:
 - View
 - Edit
 - Rename/Move
 - Download
 - Copy
 - Paste
 - (Un)Select All
 - Delete

Hands on: Creating and editing text files

You can always create files on your personal computer and transfer them to Discovery, but it is typically more efficient to create them directly on Discovery. Try using the GUI to create a script and change the editor mode to color the code and make edition easier.



Running jobs

A job consists of all the data, commands, scripts, and programs that will be used to obtain results. Because the Discovery computing cluster is a shared system, we use a **job scheduler** to manage all requests for resources. The Slurm (Simple Linux Utility for Resource Management) job scheduler is an open-source job scheduler that allocates compute resources on clusters for queued, user-defined jobs. It performs the following functions:

- Schedules user-submitted jobs
- Allocates user-requested compute resources
- Processes user-submitted jobs

[Running Jobs user guide](#)

Hands on: Run Hello World and check results

USC CARC Laguna / Job Composer

Jobs

Templates

Help

Jobs

+ New Job

☆ Create Template

Edit Files

Job Options

Open Terminal

Submit

Stop

Delete

Show 25 entries

Search:

Created	Name	ID	Cluster	Status
May 14, 2025 10:35pm	Single-core Python Job	5290	Regional Cluster	Completed

Showing 1 to 1 of 1 entries

Previous 1 Next

Job Details

5290

Job Name:

Single-core Python Job

Submit to:

Regional Cluster

Account:

Not specified

Script location:

/home1/danperez@cmc.edu/ondemand/data/sys/myjobs/projects/default/1

Script name:

job.sh

Folder Contents:

job.sh

script.py

slurm-5290.out

Submit Script

job.sh

Script contents:

```
#!/bin/bash

#SBATCH --partition=compute
#SBATCH --nodes=1
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=1
#SBATCH --mem=1G
#SBATCH --time=1:00:00

module purge
module load gcc/13.3.0
module load python/3.11.9

python script.py
```

Open Editor

Open Terminal

Open Dir

Logging in

To log in to the Laguna login node (also known as the *head node*), you will need to use a secure shell client. This is a small application that enables you to connect to a remote computer via SSH (**S**ecure **S**hell), a cryptographic network protocol for securely operating network services.

You will need your USC NetID to SSH into the login node and an SSH key assigned to your account and validated by CARC.

Hands on: SSH login

To connect, open a new terminal window and enter:

```
ssh <username>@laguna.carc.usc.edu
```

File operations

- Create
- View
- Edit
- Rename/Move
- Download
- Copy
- Delete

Creating and editing text files

You can edit text files in the shell console. For plain-text files, you can use the micro, nano, vim, or emacs text editors. We recommend the Micro editor for new users; Vim and Emacs both have steeper learning curves, but you may eventually find them more useful and productive. By default, only vi is installed on the HPC.

Copy files to project folder

For this command to work you must exit the ssh and work from your local computer

```
$scp local_file_path  
<username>@laguna.carc.usc.edu:<  
destination_path>
```

Installing and running software

- On Discovery, CARC maintains software, compilers, and libraries using the module system. These may satisfy your computing requirements without any further installations.
- To see the list of available software, enter the command:

```
$ module avail
```

- To load software based on this list, such as Julia, enter the command:

```
$ module load julia
```

Creating and submitting Slurm job scripts Cheat Sheet

- <https://www.carc.usc.edu/user-guides/hpc-systems/using-our-hpc-systems/slurm-cheatsheet>

Hands On: Parallel GIS: Zonal Statistics with MPI

High Performance Computing for Geospatial
Data

Vanessa Casillas | SP'2025 | HPC Workshop

Download a repository

```
$git clone https://github.com/
```

Why HPC for GIS?

- Geospatial datasets (like satellite imagery, DEMs) are large
- Operations like zonal statistics can be slow
- HPC enables:
 - Faster processing
 - Larger coverage areas
 - Efficient memory use

Zonal Statistics Project

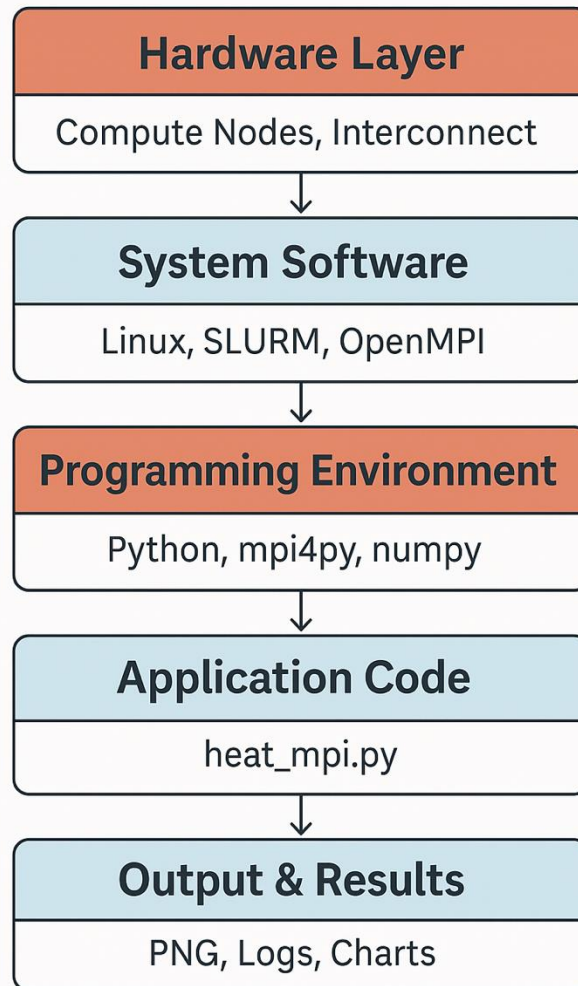
- Objective: Compute mean elevation from a DEM (GeoTIFF)
- Use mpi4py to parallelize over multiple processors
- Each process handles a chunk of the raster
- Results combined at the end
- Input: dem.tif | Output: Average Elevation (meters)

Zonal Statistics Explained

- Zonal statistics: Calculate metrics (mean, min, max) for each zone in a raster
- In this case: the entire raster = one zone
- Common in:
 - Land use analysis
 - Environmental modeling
 - Urban planning

HPC Technology Stack

HPC Technology Stack



Hands On

Create a session

```
salloc --partition=gpu --gres=gpu:1 --cpus-per-task=8 --mem=32GB  
--time=1:00:00
```

Load Conda Module

```
Module purge
```

```
Module load conda
```

[Conda](#) is the package and environment manager that installs, runs, and updates packages and their dependencies.

Initialize shell to use Conda and Mamba

```
mamba init bash
```

```
source ~/.bashrc
```

Create a virtual environment

```
mamba create --name <env_name>
```

Check environments info

```
conda info --envs
```

Activate the environment

```
Mamba activate <env name>
```

Install packages into the environment

```
python -m ipykernel install --user --name envname -  
-display-name "disp_name" -> Kernel for Jupyter lab
```

```
mamba install pytorch torchvision torchaudio pytorch-cuda=11.8 -c  
pytorch -c nvidia
```

Type 'python' and verify the packages by importing them

```
Import torch
```

Hands On: Create a project environment and Install Modules

- Python
- mpi4py – parallel processing
- rasterio – reading GeoTIFF files
- numpy – data handling
- SLURM or mpiexec – job scheduler (HPC)
- Easy to reproduce and modify for other GIS tasks

Parallel Processing Strategy

- Load raster on root process
- Divide rows equally among MPI processes
- Each process computes local mean
- Use MPI.Reduce to compute global average

Code Highlight – MPI Logic

- `rows_per_proc = shape[0] // size`
- `start = rank * rows_per_proc`
- `end = (rank + 1) * rows_per_proc`
- `local_data = data[start:end, :]`
- `local_mean = np.mean(local_data[local_data != nodata])`
- `global_mean = comm.reduce(local_mean, op=MPI.SUM, root=0)`

Running on HPC

- Script: `zonal_mpi.py`

- Run with:

```
mpiexec -n 4 python zonal_mpi.py
```

- Or via SLURM:

```
sbatch run_zonal.sh
```

```
Queue -u $USER
```

Run Batch and get Results

Job status

```
queue -u $USER
```

- Check Output file:

Cat jobid.out

Average Elevation: 543.27 meters

Key Takeaways

- HPC is powerful for processing big GIS data
- mpi4py makes parallelization accessible in Python
- Raster-based zonal statistics scale easily with more cores
- Can extend to NDVI, land cover, or terrain classification

Resources

<https://www.carc.usc.edu/user-guides/carc-ondemand>

<https://uschpc.github.io/regional-computing-website/user-guides/get-started-laguna/logging-in.html>

Additional resources

[OnDemand documentation](#)